

Shor's Algorithm

Why it works, how it is built, and what it costs

RSA and elliptic curves, from the spectral theorem down to the Toffoli count

Research seminar • 60 minutes

Three layers: Hadamards • a ladder of controlled modular multiplications • an inverse Fourier transform.

The first and third are drawn everywhere. The middle one is a box — and it is over 99% of the circuit.

	$ 1\rangle$	$ 7\rangle$	$ 4\rangle$	$ 13\rangle$
$ 0\rangle$	1	1	1	1
$ 64\rangle$	1	$-i$	-1	$+i$
$ 128\rangle$	1	-1	1	-1
$ 192\rangle$	1	$+i$	-1	$-i$

(all entries $\times \frac{1}{4}$)

This is the state of a quantum computer, just before it factors 15.

Read it by columns: each column is a phase ramp. Its slope is the answer.

Read it by rows: each row is an eigenvector of “multiply by 7 mod 15”.

Read it as a whole: it is the 4×4 discrete Fourier matrix.

Those are not three facts. They are one fact, and the whole hour is spent earning the right to say it.

Shor’s algorithm never computes the secret.

It changes basis until the secret is the label of a measurement outcome.

RSA — factoring

Given $N = pq$ with p, q prime, find p .

Best classical: number field sieve, $\exp(O(n^{1/3} \log^{2/3} n))$.

ECDLP — discrete logarithm

Given $E/\mathbb{F}_p$, a point P of prime order r , and $Q_{\text{pub}} = [k]P$, find k .

Best classical: Pollard rho, $O(\sqrt{r})$ — *generic*, so it is believed optimal against a black-box group.

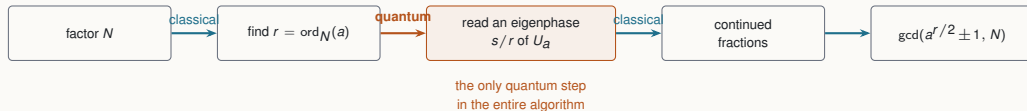
Both are instances of one statement:

Abelian hidden subgroup:
 $f : G \rightarrow S$ is constant on cosets of an
 unknown $H \leq G$
 and distinct across them. Find H .

G	H hidden by f
$\mathbb{Z}$	$r\mathbb{Z}$, from $f(x) = a^x \bmod N$
$\mathbb{Z}_r \times \mathbb{Z}_r$	the line $\langle (-k, 1) \rangle$, from $f(u, v) = [u]P + [v]Q$

A period is a *rank-one* phenomenon; a discrete logarithm is *rank two* — the secret is the *slope of a line* in a discrete torus.

RSA, finite-field Diffie–Hellman, DSA, ECDH, ECDSA: one theorem applied five times, not five coincidences.



WHAT SHOR PROVED (1994)

N can be factored, and discrete logarithms computed, in $O(n^2 \log n \log \log n)$ quantum steps and $O(n)$ qubits — *polynomial*, against superpolynomial classically.

Three things to hold onto:

- the quantum computer solves **one** subproblem;
- that subproblem is **spectral** — it reads an eigenvalue;
- everything else is nineteenth-century number theory.

- F1 States.** k qubits: a unit vector $|\psi\rangle = \sum_x \psi_x |x\rangle$ in $(\mathbb{C}^2)^{\otimes k} \cong \mathbb{C}^{2^k}$. A table of amplitudes. *Global* phase is invisible; *relative* phase is the entire resource.
- F2 Evolution.** A gate is a unitary: $U^\dagger U = 1$. So a circuit only ever *rotates* the state vector. Unitary $\Rightarrow$ invertible $\Rightarrow$ every subroutine is reversible.
- F3 Composition.** Registers combine by $\otimes$: dimensions *multiply*. $t + n$ qubits carry 2^{t+n} amplitudes. (Had nature chosen $\oplus$ there would be no subject.)
- F4 Measurement = sample, then condition.** Write $|\Psi\rangle = \sum_b \nu_b |b\rangle_A |\psi_b\rangle_B$ with $\nu_b \geq 0$: outcome b with probability ν_b^2 , state becomes $|b\rangle_A |\psi_b\rangle_B$. Collapse is conditional probability with amplitudes upstream of it.
- F5 Interference — a theorem, not a postulate.**

$$\Pr[x] = \left| \sum_j A_j \right|^2 = \underbrace{\sum_j |A_j|^2}_{\text{classical}} + \underbrace{\sum_{j \neq j'} \overline{A_{j'}} A_j}_{\text{cross terms}}$$

The cross terms can be negative. **This is the only reason quantum beats classical**, and protecting these terms is what most of Shor's circuit is *for*.

EIGENVALUES ARE ANGLES

If $U|u\rangle = \lambda|u\rangle$ with U unitary, $\|u\| = 1$, then $|\lambda| = 1$, so $\lambda = e^{2\pi i\varphi}$.

Proof. $1 = \langle u|u\rangle = \langle u|U^\dagger U|u\rangle = |\lambda|^2$. □

Moral

If a unitary is going to tell us a number, that number **must** be encoded as a phase. This is forced, not chosen.

DISTINCT EIGENVALUES $\Rightarrow$ ORTHOGONAL

$U|u\rangle = \lambda|u\rangle$, $U|u'\rangle = \lambda'|u'\rangle$, $\lambda \neq \lambda' \Rightarrow \langle u'|u\rangle = 0$.

Proof. $\langle u'|u\rangle = \langle u'|U^\dagger U|u\rangle = \overline{\lambda'}\lambda \langle u'|u\rangle$ and $\overline{\lambda'}\lambda \neq 1$. □

Moral

This is why, later, the r eigenbranches of Shor's state can **never** interfere with one another — and why one measurement cleanly picks one of them.

Add the spectral theorem — every unitary has an orthonormal eigenbasis, $U = \sum_j e^{2\pi i\varphi_j} |u_j\rangle\langle u_j|$ — and Part I is complete.

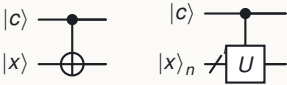
The single most misleading thing

The horizontal lines are **not wires** and **nothing flows along them**. Each line is one qubit; moving right is moving forward in time.

It is closer to **sheet music** than to a plumbing schematic: one staff line per qubit, read left to right, a column is one instant.

So a diagram with eight lines has eight qubits and always will. Lines never split, merge, or get created. **Read the two edges first** — the far left is the input, the far right the output. The middle is only the recipe.

symbol	meaning
•	a <i>control</i> : look, change nothing
⊕	<i>target</i> : flip if every dot is 1
box	one joint operation on those qubits
vertical line	not a wire — “simultaneous”
/ <i>n</i>	a bundle of <i>n</i> qubits



The smallest example already contains all of Shor:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ in } \{|0\rangle, |1\rangle\}, \quad X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ in } \{|+\rangle, |-\rangle\}, \quad X = HZH.$$

In the computational basis X is a **permutation**: no entry is complex — you cannot see a phase anywhere.

In its own eigenbasis the same operator **moves nothing**: it fixes $|+\rangle$ and multiplies $|-\rangle$ by $-1 = e^{i\pi}$.

One rotation, two descriptions. H is the dictionary.

One size up: in $\{|0\rangle, |1\rangle\} \otimes \{|+\rangle, |-\rangle\}$,

$$\text{CNOT} = \text{diag}(1, 1, 1, -1).$$

A *controlled phase* — and a diagonal gate has **no preferred control wire** ($e^{ixy\phi}$ is symmetric in $x \leftrightarrow y$), which is why such gates are drawn as two dots joined by a line.

That is phase kickback, one section early.

Moral

Every unitary owns a basis — its eigenbasis — in which nothing mixes and each axis merely spins by a phase. **Pick the basis that diagonalises the operator you care about, and mysteries become bookkeeping.** “Which register acts on which” is a property of the description, not of the operator.

Why it works

Fix N odd, composite, not a prime power. Pick a with $\gcd(a, N) = 1$; let $r = \text{ord}_N(a)$.

THE SPLITTING THEOREM

If r is *even* and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2} - 1, N)$ is a nontrivial factor of N .

Proof. Put $x = a^{r/2}$, so $N \mid x^2 - 1 = (x - 1)(x + 1)$. $N \nmid x - 1$ (else r was not minimal) and $N \nmid x + 1$ (hypothesis). So the primes of N *split* between the two factors: $1 < \gcd(x - 1, N) < N$. $\square$

THE HALF THEOREM

For odd N with $m \geq 2$ distinct prime factors and a uniform on the units, the hypotheses hold with probability $\geq 1 - 2^{-(m-1)} \geq \frac{1}{2}$.

(CRT + counting 2-adic valuations of the local orders. The exponent is $m - 1$, not m : at $N = 21$ exactly half the good bases work.)

Note the direction: the bound *improves* with the number of prime factors — so the hardest case for the classical stage is exactly $N = pq$, the RSA case.

Budget for failure: $N = 33$, exhaustively

Of the 19 nontrivial bases coprime to 33:

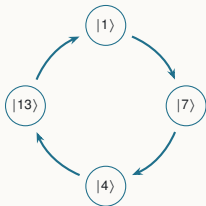
r odd ($r = 5$)	$a = 4, 16, 25, 31$
$a^{r/2} \equiv -1$	$a = 2, 8, 17, 29, 32$
succeed	10 of 19

47% of bases send you back to the top of the loop with a *correct* order in hand. Order finding worked; the reduction failed. This is the common case, not an error.

On n qubits define

$$U_a |y\rangle = |ay \bmod N\rangle \quad (y < N), \quad U_a |y\rangle = |y\rangle \quad (N \leq y < 2^n).$$

$\gcd(a, N) = 1$ makes $y \mapsto ay$ a *bijection*, so U_a is a permutation matrix, hence unitary. *Unitarity here is a number-theoretic condition.*



$$N = 15, \quad a = 7, \quad r = 4$$

On the orbit $\{1, a, a^2, \dots, a^{r-1}\}$, U_a is a **single r -cycle**. The period is present in the circuit as *the length of a cycle*. Two matrices, one operator ($r = 4$):

$$[U_a]_{\text{comp}} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad [U_a]_{\text{eig}} = \begin{pmatrix} 1 & & & \\ & i & & \\ & & -1 & \\ & & & -i \end{pmatrix}$$

Every entry on the left is 0 or 1: r is invisible in the matrix entries. It is written plainly in the eigenvalues.

Phase estimation is the instrument that reads eigenvalues.

NECKLACE LEMMA

On the orbit of 1 the eigenvalues of U_a are exactly $e^{2\pi is/r}$, $s = 0, \dots, r-1$, each simple, with orthonormal eigenvectors

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle.$$

Proof. Try $|f\rangle = \sum_k f(k) |a^k\rangle$ (indices mod r). Then $U_a |f\rangle = \sum_k f(k-1) |a^k\rangle$, so $U_a |f\rangle = \lambda |f\rangle$ iff $f(k-1) = \lambda f(k)$, i.e. $f(k) = \lambda^{-k} f(0)$. Consistency round the loop, $f(k+r) = f(k)$, forces $\lambda^r = 1$. Orthonormality is the character sum $\frac{1}{r} \sum_k e^{2\pi i (s'-s)k/r} = \delta_{ss'}$; and r orthonormal vectors in an r -dimensional space leave no room for more. $\square$

Moral

U_a is a necklace shift. **Shifts are diagonalised by waves.** The wave with winding number s has eigenphase s/r — so r sits in every dial speed.

Why the phases in $|u_s\rangle$ are *forced*: the unphased orbit sum has eigenvalue 1, because the shift merely permutes identical terms. One must break the symmetry by attaching phases — and then the eigenvalue picks up exactly one factor $e^{2\pi is/r}$: a number containing r .

Writing down $|u_s\rangle$ requires knowing r — the very thing we seek. Phase estimation looks circular.

DELTA-FLAT LEMMA

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle, \quad \text{and more generally} \quad |a^k \bmod N\rangle = \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} e^{2\pi i s k / r} |u_s\rangle.$$

Proof. Exchange the sums: $r^{-1/2} \sum_s |u_s\rangle = r^{-1} \sum_k (\sum_s e^{-2\pi i s k / r}) |a^k\rangle$, and the inner geometric series is $r \delta_{k,0}$. $\square$

Moral

The Fourier transform of a delta function is flat, transported to the orbit. $|1\rangle = |0 \cdots 01\rangle$ costs **one X gate**, and it is secretly the uniform superposition of all r eigenvectors — none of which we could write down.

The circuit is linear, so running it on $|1\rangle$ runs it on *every* eigenbranch at once. The branches are orthogonal (Lemma 2), so they cannot interfere; measurement conditions onto one.

Price: we get s/r for a uniformly random s , *not chosen by us*. Continued fractions pay that price.

$$c\text{-}U = |0\rangle\langle 0| \otimes \mathbb{I} + |1\rangle\langle 1| \otimes U = \begin{pmatrix} 1 & 0 \\ 0 & U \end{pmatrix}$$

KICKBACK

If $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ then for *any* control amplitudes $c\text{-}U[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle] = (\alpha|0\rangle + \beta e^{2\pi i\varphi}|1\rangle) \otimes |u\rangle$.

Proof. The state maps to $\alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes e^{2\pi i\varphi}|u\rangle$. Move the scalar across the tensor sign — legitimate because $\otimes$ is bilinear, $|a\rangle \otimes (\lambda|b\rangle) = (\lambda|a\rangle) \otimes |b\rangle$ — and factor out $|u\rangle$. □

The proof is three lines and consists **entirely of moving a scalar across a tensor sign**. Nothing physically travelled. Three things happened at once, and they are the crux:

(i) The target does not move. An eigenvector is a fixed point up to a scalar. The work register is a *catalyst*: it enters $|u\rangle$, leaves $|u\rangle$, and — no-cloning notwithstanding — one copy serves every gate in the circuit.

(ii) The scalar lands on the control. What would have been an *unobservable global* phase is now a *relative* phase between control basis states. Information flowed backwards, against the way the gate points.

(iii) The registers stay a product. No entanglement is created. Kickback is *not* an entangling phenomenon; eigenvectors are exactly the states on which a controlled gate is coherence-preserving.

Expand $U = \sum_j e^{2\pi i \varphi_j} |u_j\rangle\langle u_j|$ inside c- U :

$$\text{c-}U = |0\rangle\langle 0| \otimes \mathbb{I} + \sum_j e^{2\pi i \varphi_j} |1\rangle\langle 1| \otimes |u_j\rangle\langle u_j| \implies [\text{c-}U]_{\{|c\rangle \otimes |u_j\rangle\}} = \text{diag}(1, \dots, 1, e^{2\pi i \varphi_1}, e^{2\pi i \varphi_2}, \dots)$$

In the eigenbasis of the target, controlled- U is a conditional phase. Nothing else.

The smallest instance is the CNOT again:

$$\text{CNOT } |+\rangle |-\rangle = |-\rangle |-\rangle,$$

a *bit flip on the control*, produced by a gate that “acts on the target”. In the computational basis this looks like magic; in $\{|0\rangle, |1\rangle\} \otimes \{|\pm\rangle\}$ it is $\text{diag}(1, 1, 1, -1)$ doing the only thing diagonal matrices do. **Shor’s $|u_s\rangle$ is the same trick with a finer protractor.**

If the target is not an eigenvector

$\text{c-}U |+\rangle |\psi\rangle = \frac{1}{\sqrt{2}} \sum_j c_j (|0\rangle + e^{2\pi i \varphi_j} |1\rangle) |u_j\rangle$: each eigencomponent kicks back *its own* phase, so the control state depends on which component it is glued to. Rank of the coefficient matrix > 1 : **the registers entangle**, and the control’s interference fringes fall to visibility $|z|$, $z = \langle \psi | U | \psi \rangle$.

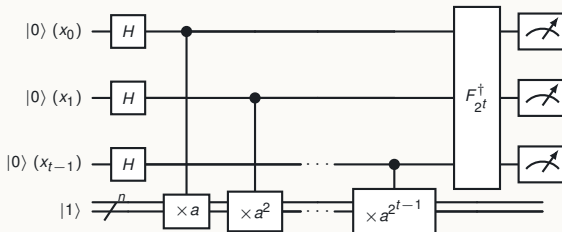
SHIFT THEOREM

Let $A_c |y\rangle = |(y + c) \bmod M\rangle$ on $\mathbb{Z}_M$. The Fourier states $|\hat{\ell}\rangle = F_M^\dagger |\ell\rangle$ are simultaneous eigenvectors of every shift: $A_c |\hat{\ell}\rangle = e^{2\pi i c \ell / M} |\hat{\ell}\rangle$, i.e. $F_M^\dagger A_c F_M$ is diagonal.

M	where	what it gives you
2	$A_1 = X, F_2 = H, \hat{1}\rangle = -\rangle$	the CNOT/ $ -\rangle$ kickback; the textbook oracle trick $U_f x\rangle -\rangle = (-1)^{f(x)} x\rangle -\rangle$
2^n	addition of a constant	run in reverse: builds an adder out of phase gates (Draper)
r	U_a on the orbit of 1	the spectrum $e^{2\pi i s / r}$ and the eigenvectors $ u_s\rangle$ — Shor
2^t	the counting register at the end	$F^\dagger$ rotates the phase ramp back onto the computational basis

Moral

Four uses of one linear-algebra fact at four different scales. For a mathematician the compact statement is: $\hat{G} \cong G$ for finite abelian G , the characters are an orthonormal basis, and $F_G : \chi_c / \sqrt{|G|} \mapsto \delta_c$ is the change of basis from frequencies to positions. Every algorithm here is an arrangement for making the answer a position in the basis you are about to measure.



Rung j applies U^{2^j} conditioned on bit x_j , and $\prod_j (U^{2^j})^{x_j} = U^x$. On an eigenvector, branch $|x\rangle$ acquires $e^{2\pi i x \varphi}$ and $|u\rangle$ factors out:

$$\frac{1}{\sqrt{2^t}} \sum_x |x\rangle \otimes |u\rangle \mapsto \underbrace{\left[\frac{1}{\sqrt{2^t}} \sum_{x=0}^{2^t-1} e^{2\pi i \varphi x} |x\rangle \right]}_{\text{a phase ramp of frequency } \varphi} \otimes |u\rangle \quad (\text{still a product state})$$

Moral

As x increases by 1 the phase advances by $2\pi\varphi$: a discrete plane wave, one bit of φ per control qubit. **Measuring the frequency of a wave is what Fourier transforms do.**



needle angle = $2\pi\varphi x$: a helix of pitch φ

$F^\dagger$ is a pitch-meter

The problem nobody states

The ladder's rungs are $U, U^2, U^4, \dots, U^{2^{t-1}}$. Building them by repetition costs $1 + 2 + 4 + \dots + 2^{t-1} = 2^t - 1$ applications of U : **exponentially many**.

Phase estimation is efficient only for unitaries whose powers are cheap. For a generic black box, or a Hamiltonian evolution e^{iHt} , there is no shortcut and the procedure offers no speedup at all.

Moral

[Gift 1] **The powers collapse.** $U_a^{2^j} = U_{a^{2^j \bmod N}}$.

A high power of U_a is not a long circuit — it is *the same circuit with a different classical constant*, precomputed by repeated squaring before the machine is switched on. $2^t - 1$ applications become t rungs.

Moral

[Gift 2] **One rung is cheap.** U_a itself has a circuit of size $O(n^2)$.

This is the entire second half of the talk — reversible modular arithmetic.

Gift 1 without Gift 2: a short ladder of unaffordable rungs. Gift 2 without Gift 1: an affordable rung repeated an unaffordable number of times.

Shor's algorithm exists in the narrow space where arithmetic hands you both.

A ramp of frequency $\ell/2^t$ is the Fourier basis vector $F_{2^t} |\ell\rangle$. So “measure in the Fourier basis” = “apply $F^\dagger$, then measure”. The amplitude of outcome ℓ is

$$a_\ell = \frac{1}{2^t} \sum_{x=0}^{2^t-1} e^{2\pi i x \delta}, \quad \delta = \varphi - \frac{\ell}{2^t}.$$

Each ℓ is the *hypothesis* “the frequency is $\ell/2^t$ ”, and $F^\dagger$ ’s factor $e^{-2\pi i x \ell/2^t}$ unwinds the ramp at that trial rate.

Right hypothesis: every term is 1 — 2^t unit vectors adding coherently.

Wrong hypothesis: the residual phases sweep the circle and cancel.

Constructive interference on the truth, destructive everywhere else.

$|a_\ell|^2$ is a Dirichlet-kernel peak of width ~ 1 centred on $2^t \varphi$, with $1/\ell^2$ tails; each extra counting qubit buys one more bit, additively. **This is a bandwidth limit, not a quantum one:** a finite window cannot resolve a frequency exactly.

QPE ACCURACY

Let ℓ be nearest to $2^t \varphi$, so $|2^t \delta| \leq \frac{1}{2}$. Then

$$\Pr[\ell] = |a_\ell|^2 \geq \frac{4}{\pi^2} \approx 0.405,$$

and the two nearest integers carry $\geq 8/\pi^2 \approx 0.81$.

Proof. $|a_\ell| = 2^{-t} |\sin(\pi 2^t \delta) / \sin(\pi \delta)|$; bound the numerator below by the chord ($|\sin \pi \theta| \geq 2|\theta|$) and the denominator above by the arc ($|\sin u| \leq |u|$). □

Reading A — computational basis

The ladder tabulates f :

$$\frac{1}{\sqrt{2^t}} \sum_x |x\rangle |1\rangle \mapsto \frac{1}{\sqrt{2^t}} \sum_x |x\rangle |a^x \bmod N\rangle$$

Literally a **lookup table** of $f(x) = a^x \bmod N$: each branch is a row, and $|x\rangle$ is not consumed — it is the *graph* of f , all 2^t rows at once.

Shor, 1994: build a periodic function, Fourier-sample its period.

Reading B — eigenbasis

Substitute $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$ and swap the sums:

$$= \frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} \left[\underbrace{\frac{1}{\sqrt{2^t}} \sum_x e^{2\pi i x s / r} |x\rangle}_{\text{ramp of frequency } s/r} \right] \otimes |u_s\rangle$$

One QPE phase ramp per eigenbranch, tagged by an orthogonal label.

Kitaev; Cleve–Ekert–Macchiavello–Mosca: phase estimation on U_a .

No gate ever writes a phase

Every gate in the ladder is a **permutation matrix**: below the final $F^\dagger$ there is no complex number anywhere in the circuit. The phases in Reading B are not *added* by anything — they are what a permutation *looks like* in its own Fourier eigenbasis, and they become physically meaningful only because the permutation was applied to a *superposition*. **The phases appear on the change of basis — and that change of basis is the algorithm.**

Why one run yields one eigenphase

[22:00]

Apply $F^\dagger$ to the counting register of Reading B. Because the tags $|u_s\rangle$ are mutually orthogonal, the cross terms die and

$$\Pr[\ell] = \left\| \sum_s \frac{1}{\sqrt{r}} A_\ell(s/r) |u_s\rangle \right\|^2 = \frac{1}{r} \sum_{s,s'} \overline{A_\ell(s'/r)} A_\ell(s/r) \underbrace{\langle u_{s'} | u_s \rangle}_{\delta_{ss'}} = \frac{1}{r} \sum_s |A_\ell(s/r)|^2.$$

SAMPLING THEOREM

The statistics are *exactly* those of: **draw s uniformly from $\{0, \dots, r-1\}$, then run ideal phase estimation on s/r .**

Moral

Linearity superposes the histories; orthogonality *forbids* their interference; the Born rule then samples one dial speed per run.

This is an equality of *statistics*, not a hidden classical coin — the circuit stays coherent throughout. The work register is a perfect, unread which-branch record, so the r ramps coexist without blurring. **Eigenphases are sampled, never averaged.**

A common misreading

The pre-measurement state contains all r eigenphases — but this is **not** “computing all eigenvalues at once”. One measurement returns *one*, with probability $1/r$.

What makes Shor work is not massive parallelism but that **any single good sample suffices**, and that interference concentrated the probability on the good outcomes *before* the sample was taken.

For finite abelian G : a character is a homomorphism $\omega : G \rightarrow \mathbb{C}^\times$; the characters form $\widehat{G} \cong G$; and

$$F_G = \frac{1}{\sqrt{|G|}} \sum_{x \in G} \sum_{y \in \widehat{G}} \omega_y(x) |y\rangle \langle x|$$

is unitary *exactly because* characters are orthogonal.

The standard recipe. Prepare $\frac{1}{\sqrt{|G|}} \sum_x |x\rangle |f(x)\rangle$; discard the second register; apply F_G ; measure. The output is **uniform on the annihilator**

$$H^\perp = \{y : \omega_y(h) = 1 \ \forall h \in H\}, \quad |H^\perp| = |G|/|H|.$$

Since $H = (H^\perp)^\perp$, sampling $H^\perp$ determines H .

Moral

The unknown coset representative contributes only an overall phase $\omega_y(c)$ — and therefore **nothing to** $|\cdot|^2$. That single observation is why the second register need never be measured.

Why abelian, precisely

For abelian G every irrep is one-dimensional (Schur), so “irrep” = “character” and F_G is *scalar*-valued. With irreps of dimension > 1 the transform is *matrix*-valued, measurement returns a representation label plus matrix indices, and reconstructing H appears to need entangled measurements across many coset states.

This is the whole reason lattice cryptography is believed safe.

With probability $\geq 4/\pi^2$ the measurement returns an ℓ with $|\frac{\ell}{2^t} - \frac{s}{r}| \leq 2^{-(t+1)}$.

LEGENDRE'S CRITERION

If $|\xi - \frac{p}{q}| < \frac{1}{2q^2}$ with $\gcd(p, q) = 1$, then p/q is a convergent of the continued-fraction expansion of ξ .

Since $r < N$, it suffices that $2^{-(t+1)} \leq \frac{1}{2N^2} < \frac{1}{2r^2}$, i.e.
 $2^t \geq N^2$: **take** $t = 2n$.

The register size is imposed by the *classical post-processing*.
 The quantum circuit does not care how large t is; each extra qubit is one more rung.

Then: expand $\ell/2^t$ by the Euclidean algorithm, list the $O(n)$ convergents, take the largest denominator $q < N$, verify $a^q \equiv 1$.

In code this whole step is one line: `Fraction(y, 2**t).limit_denominator(N-1).denominator` — Python ships the number theory.

Three things must go right

$$\Pr[\text{usable}] \geq \underbrace{\frac{4}{\pi^2}}_{\text{peak resolved}} \cdot \underbrace{\frac{\phi(r)}{r}}_{\gcd(s,r)=1} \cdot \underbrace{\left(1 - \frac{1}{N}\right)}_{\text{window}}$$

$\phi(r)/r = \Omega(1/\log \log r)$ — savage in asymptotic notation, negligible in practice: the Rosser–Schoenfeld bound is still **0.135** at $r = 2^{64}$.
 Or take lcm of two runs' denominators: correct with probability $\geq 6/\pi^2$.

Orbit $1 \rightarrow 7 \rightarrow 4 \rightarrow 13 \rightarrow 1$, so $r = 4$ and $r \mid 2^t$: the ramps are *exact* Fourier states. Grouping the 256 terms by work value and applying $F^\dagger$ gives $|\Psi\rangle$ described entirely by the array of the first slide, and

$$\Pr[\ell] = \frac{1}{4} \text{ for } \ell \in \{0, 64, 128, 192\}, \quad 0 \text{ else.}$$

Verified by direct simulation.

Say $\ell = 192$: $\frac{192}{256} = \frac{3}{4} = [0; 1, 3]$, convergents $0, 1, \frac{3}{4}$, so $r = 4$; check $7^4 = 2401 \equiv 1$. Then $7^2 \equiv 4 \not\equiv -1$, and

$$\gcd(3, 15) = 3, \quad \gcd(5, 15) = 5.$$

$\ell = 128$ gives $\frac{1}{2}$: candidate 2, rejected by the verifier.
 $\phi(4)/4 = \frac{1}{2}$, live.

$N = 15$ is misleadingly easy

$r = 4$ is a power of two, so $r \mid 2^t$: **no leakage, no Dirichlet tails, and continued fractions never earn their keep.**

Worse, the ladder constants $a^{2^j} \bmod 15$ are $7, 4, 1, 1, 1, \dots$ — **most of the ladder is literally the identity.** That is why 15 is everyone's demo integer.

$N = 21, a = 2$ is the honest test ($r = 6 \nmid 2^t$)

s	0	1	2	3	4	5
peak ℓ	0	171	341	512	683	853
$\ell/2^{10} \rightarrow$	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{2}{3}$	$\frac{5}{6}$
r recovered	1	6	3	2	3	6
verifier	✗	✓	✗	✗	✗	✓

Two of six — that is $\phi(6)/6$. Only **78.9%** of the mass sits in the main lobes, and that fraction *does not improve* with more qubits. **In the measured run the most frequent nonzero outcome was 512, giving $r = 2$: trusting the top outcome fails.**

The obstruction: garbage

Let $|\Psi\rangle = \sum_x \alpha_x |x\rangle_A |g_x\rangle_B$ be the state before the final basis change (A : what we transform and measure; B : *everything else* — scratch, work, environment). Apply V to A alone; by F4 the outcome probability is the squared norm of the whole y -component:

The master formula — one damping factor per unmeasured register

$$P(y) = \left\| \sum_x V_{yx} \alpha_x |g_x\rangle \right\|^2 = \sum_{x,x'} \overline{V_{yx'} \alpha_{x'}} V_{yx} \alpha_x \langle g_{x'} | g_x \rangle$$

The register you are *not* measuring casts its vote: every cross term is multiplied by the overlap of the records.

Moral

[Case 1: no record] $|g_x\rangle = |g\rangle$ for all x ; it factors out.

$$P(y) = \left| \sum_x V_{yx} \alpha_x \right|^2$$

The sum sits **inside** the modulus. Cross terms alive, full interference.

Stated exactly: amplitudes belong to *complete configurations of the whole machine*, so two branches share an amplitude — and can add or cancel — precisely when they agree on *every* register.

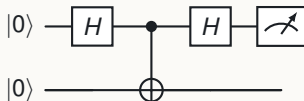
The tax

A perfect record: $\langle g_{x'} | g_x \rangle = \delta_{xx'}$.

$$P(y) = \sum_x |V_{yx} \alpha_x|^2$$

The sum has moved **outside** the modulus.

Probabilities add where amplitudes should have. In between, the fringe visibility is exactly $|\langle g_{x'} | g_x \rangle|$.



Clean ($g_0 = g_1$): $P(0) = \left|\frac{1}{2} + \frac{1}{2}\right|^2 = 1$, $P(1) = \left|\frac{1}{2} - \frac{1}{2}\right|^2 = 0$

Copied ($\langle g_1 | g_0 \rangle = 0$): $P(0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2} = P(1)$

$$\alpha_0 = \alpha_1 = \frac{1}{\sqrt{2}}, V = H, \text{ so } c_x = V_{yx}\alpha_x = \pm \frac{1}{2}.$$

The tax

Nothing was measured and nothing was lost. One CNOT onto a helper qubit made the branches disagree *somewhere*, and that was enough.

This is the double slit with a which-path detector: the fringes vanish **even if nobody ever reads the detector**, because interference requires the two paths to end in the same total state of the world.
Visibility V and distinguishability D obey $V^2 + D^2 \leq 1$ — complementarity, exactly.

Branches interfere to the extent that the rest of the universe cannot distinguish them.

Reversible arithmetic built naively computes $|x\rangle |1\rangle |0 \dots 0\rangle \mapsto |x\rangle |a^x\rangle |g_x\rangle$, where g_x collects carry chains, comparison bits, partial products — and depends on x , because the carries depend on what was added. Then

$$P(\ell) \propto \sum_{x,x'} e^{2\pi i(x-x')\ell/2^t} \underbrace{\langle a^{x'} | a^x \rangle}_{\text{keep this}} \underbrace{\langle g_{x'} | g_x \rangle}_{\text{destroy this}}$$

Moral

[The accumulator] $\langle a^{x'} | a^x \rangle$ confines interference to $x \equiv x' \pmod{r}$ — and *those* surviving cross terms are how the period imprints itself on the statistics. **It is the answer.** Which is also why the work register need never be measured.

The tax

If the scratch distinguishes x from x' *within* a residue class — and intermediate carries essentially always do — then $\langle g_{x'} | g_x \rangle = 0$, every cross term dies, and

$$P(\ell) = 2^{-t} \text{ for every } \ell : \text{ exactly uniform noise.}$$

The circuit is still perfectly unitary; it simply computes nothing.

Debugging corollary — and it is not a footnote

The failure mode of a dirty ancilla is **not a wrong answer but a flat histogram** — and a post-processor that verifies its own candidates can “succeed” on white noise, masking the bug. **Assert that ancillas end in $|0\rangle$; test the distribution, not the factors.**

$$N = pq \bullet n = \lceil \log_2 N \rceil \bullet r = \text{ord}_N(a) \bullet t = 2n \bullet U_a |y\rangle = |ay \bmod N\rangle \bullet |u_S\rangle = \frac{1}{\sqrt{r}} \sum_k e^{-2\pi i k S / r} |a^k\rangle \bullet F |j\rangle = \frac{1}{\sqrt{Q}} \sum_k e^{2\pi i j k / Q} |k\rangle$$

Three plausible alternatives, each instructively wrong:

- **Overwrite with** $|0\rangle$. Many-to-one — not unitary.
- **Discard it.** Ignoring a register *is* tracing it out; the overlap factor enters $P(y)$ regardless of whether anyone looks.
- **Measure and reset.** Projects onto the branches consistent with one value. That does not restore interference; it throws away the rest of the superposition.

Only reversibility itself provides the exit: every step that wrote the garbage is invertible, so run those steps backwards and *unwrite* it.

BENNETT: COMPUTE—COPY—UNCOMPUTE

If f has a reversible circuit $C : |x\rangle |0\rangle \mapsto |x\rangle |f(x), g_x\rangle$ of size T , then a circuit of size $2T + O(|f(x)|)$ realises

$$|x\rangle |y\rangle |0\rangle_{\text{scr}} \longmapsto |x\rangle |y \oplus f(x)\rangle |0\rangle_{\text{scr}},$$

with every ancilla returned to $|0\rangle$ **in every branch**.

Proof. Run C ; copy the answer with CNOTs (fan-out of a basis value is not cloning); run $C^\dagger$. The ancilla is now $|0\rangle$ independently of x , so it factors out of any superposition and every overlap returns to 1: Case 1 restored.

□

The tax

The price is **structural**: roughly half of every arithmetic circuit below exists to run the other half backwards. **That is not waste — it is the price of the interference.** When $F^\dagger$ finally asks the branches to recombine, nothing anywhere in the machine may remember which branch was which.

Suppose the scratch takes orthogonal values on blocks $B_1, \dots, B_\tau$ partitioning the r branches of a coset, $|B_\beta| = n_\beta$. The complete character sum over $\mathbb{Z}_r$ cancels; *a partial sum over a proper subset does not*. Hence

$$\Pr[\text{peak}] = \frac{1}{r^3} \sum_{\beta} n_{\beta}^2, \quad \frac{\text{signal on the dual line}}{\text{ideal}} = \frac{\sum_{\beta} n_{\beta}^2}{r^2} \leq 1,$$

with equality **iff** $\tau = 1$. Any record that splits a coset at all strictly reduces the signal.

leaked	blocks	peak (sim.)	signal
nothing	[7]	0.14286	1.000
parity of x_1	[4, 3]	0.07289	0.510
$x_1 \bmod 3$	[3, 2, 2]	0.04956	0.347
x_1 entirely	[1 ⁷]	0.02041	0.143

$r = 7, k = 4$: closed form against direct simulation, agreement exact.

One leaked bit halves the signal ($2(r/2)^2/r^2 = \frac{1}{2}$). A full leak gives $1/r$: the uniform distribution, zero information.

And for *partially* distinguishable records with overlap s :

$$\Pr[y] = \underbrace{\frac{s}{r} [y_2 \equiv ky_1]}_{\text{ideal}} + \underbrace{\frac{1-s}{r^2}}_{\text{white noise}}$$

Ideal distribution and white noise, mixed with weights s and $1 - s$. Linear in the overlap, with no threshold to hide behind.

Building it: reversible arithmetic and Qiskit

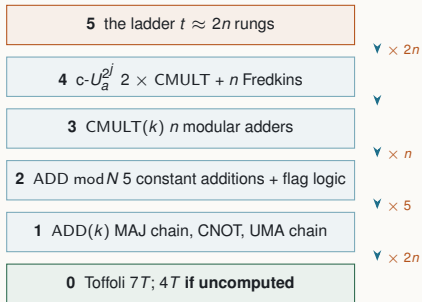
Phase estimation does **not** accept just any circuit that computes $a^y \bmod N$.

Demand	Why	Discharged by
Unitary	the whole framework is spectral; a non-unitary map has no eigenphase to read	$\gcd(a, N) = 1 \Rightarrow$ permutation
Cheap powers	else the ladder costs $2^t - 1$ applications	Gift 1: $U_a^{2^j} = U_{a^{2^j \bmod N}}$
Controllable	an uncontrolled U_a emits only a <i>global</i> phase, which no measurement can see	a control threaded through every level
Garbage-free	retained scratch entangles and kills the interference	uncomputation at every level

Moral

Unitarity here is a number-theoretic condition. $y \mapsto ay \bmod N$ is injective on $\{0, \dots, N - 1\}$ precisely when a is invertible. If that fails, the map collapses distinct inputs together, is not a permutation, is not unitary — **so there is no operator, no spectrum, and nothing for phase estimation to estimate.** (In the algorithm this is checked first, and if it fails you have already factored N by computing the gcd.)

The same hypothesis returns at Level 4 in a second role: $\bar{a} = a^{-1} \bmod N$ is what makes the multiplier realisable *in place*. **One hypothesis, two jobs.**



$$\underbrace{2n}_{\text{rungs}} \cdot \underbrace{2n}_{\text{adds}} \cdot \underbrace{5}_{\text{Toffoli}} \cdot \underbrace{2n}_{\text{Toffoli}} \approx 40n^3$$

Moral

Three of those steps each multiply by roughly n — rungs, bits of the multiplicand, bits per addition. **That, and nothing subtler, is why the algorithm is cubic.**

Windowing plus the control lemma bring $40n^3$ down to $\approx 0.3n^3$.

Every gate in this tower is a permutation matrix. The entire structure is a *classical reversible circuit* that happens to act linearly on superpositions. No complex number appears anywhere below Level 5 — which is exactly why the phases of Shor's algorithm cannot be attributed to any gate.

With $c_0 = 0$: $s_i = a_i \oplus b_i \oplus c_i$, $c_{i+1} = \text{MAJ}(a_i, b_i, c_i) = a_i b_i \oplus a_i c_i \oplus b_i c_i$.

Moral

Read it as a *cost* statement. The sum bits are $\mathbb{F}_2$ -**linear** given the carries, so they are CNOTs and essentially free. The carries are **quadratic**, so each costs a Toffoli. And they form a **chain**, so the depth is $\Theta(n)$. **Linear parts free, quadratic parts expensive, chain gives depth: that one line prices Shor's algorithm.**

Cuccaro–Draper–Kutin–Moulton, in place, one ancilla, two three-gate blocks on a wire triple (c, b, a) :

MAJ : CNOT($a \rightarrow b$), CNOT($a \rightarrow c$), TOF($c, b \rightarrow a$)

UMA : TOF($c, b \rightarrow a$), CNOT($a \rightarrow c$), CNOT($c \rightarrow b$)

CDKM: $2n$ Toffoli, $4n+1$ CNOT, $2n+2$ qubits, a preserved, in place on b , **zero garbage**; fault-tolerantly $\approx 4n T$.

```
def maj(qc, c, b, a):
    qc.cx(a, b); qc.cx(a, c); qc.ccx(c, b, a)

def uma(qc, c, b, a):
    qc.ccx(c, b, a); qc.cx(a, c); qc.cx(c, b)

def cdkm_add(qc, a, b, c0, z):
    n = len(a)
    T = [(c0, b[0], a[0])] + \
        [(a[i-1], b[i], a[i]) for i in range(1, n)]
    for t in T: maj(qc, *t)
    qc.cx(a[n-1], z) # the one clean carry
    for t in reversed(T): uma(qc, *t)
```

The idea, in one sentence

UMA erases the carry and writes the sum bit in the same three gates. Adding by hand you scribble carry digits above the columns; here **you go back over the columns rubbing out each scribble, and the act of rubbing out deposits the answer. You are not paying twice; you are paying once and collecting on the way out.**

The Shift Theorem at $M = 2^n$, read as a design principle:
adding a constant is a shift, and shifts are diagonal in the Fourier basis. The QFT factors,

$$F_{2^n} |y\rangle = \bigotimes_{i=0}^{n-1} \frac{|0\rangle + e^{2\pi i y 2^{i-n}} |1\rangle}{\sqrt{2}},$$

so a *classically known* X is added by n **uncontrolled single-qubit phase gates**: qubit i gets $P(2\pi X 2^{i-n})$. No carries, no ancillas, **addition layer of depth 1**.

Subtraction is the same gates with $-X$ — phases are periodic, so negative constants just work. And the composite $F^\dagger \cdot (\text{phases}) \cdot F$ is *exactly* the same 0/1 permutation matrix as CDKM: **two factorisations of one permutation**, one through $\mathbb{F}_2$ carry algebra, one through the Fourier dual.

Measured, same modular adder, transpiled to $\{\text{CNOT}, U\}$: at $n = 4$ Fourier uses 7 qubits / 125 rotations, ripple-carry 13 qubits / 71 Toffoli. **Fourier rotations grow as $O(n^2)$ per addition; ripple-carry Toffolis as $O(n)$.** For a laptop simulation Fourier wins on qubit count; **for real fault-tolerant hardware ripple-carry wins and it is not close.**

```
def add_const(qc, X, y):      # |y> -> |y + X mod 2^n>
    n = len(y)
    qc.append(qft(n), y)
    for i in range(n):
        qc.p(2*np.pi * X * 2.0**(i - n), y[i])
    qc.append(qft(n).inverse(), y)
```

The tax

The smallest rotation is $2\pi/2^{n+1}$:
 0.20 rad at $n = 4$; 10^{-9} at $n = 32$;

10^{-616} rad at $n = 2048$

— a number no apparatus will ever apply. Worse, arbitrary-angle rotations are **not fault-tolerant gates**; each must be synthesised from Clifford+ T .

$(x + X) \bmod N$ is classically “add; if the result is $\geq N$, subtract N ”. But the machine holds a **superposition in which some branches overshot and some did not**: there is no single answer to “did it overshoot?”. **The conditional must become arithmetic.**

	operation	value register	flag f
(1)	ADD(X)	$x + X$ ($< 2N$, fits)	0
(2)	SUB(N)	$x + X - N$; sign = $[x + X < N]$	0
(3)	CNOT sign $\rightarrow f$	unchanged	$[x + X < N]$
(4)	ADD(N) if f	$R := (x + X) \bmod N$	unchanged
(5)	SUB(X)	$R - X$; sign = $[R < X]$	unchanged
(6)	X on f ; CNOT	unchanged	0
(7)	ADD(X)	R	0

Blocks (1)–(4) compute the answer. **Blocks (5)–(7) exist entirely to destroy the evidence.** The value register is $n+1$ wide so intermediates up to $2N$ fit.

```
def c_add_mod(qc, ctrls, X, y, sign, flag, N):
    yw = list(y) + [sign] # (n+1)-wide
    c_add_const(qc, ctrls, X, yw) # (1) +X
    add_const(qc, -N, yw) # (2) -N
    qc.cx(ctrls, flag) # (3)
    c_add_const(qc, [flag], N, yw) # (4) +N if flag
    add_const(qc, -X, yw) # (5) -X
    qc.mcx(ctrls + [flag], flag) # (6) clear flag
    qc.x(flag)
    add_const(qc, X, yw) # (7) +X
```

Deduce the flag, don't remember it

After (4) you hold the right answer but a qubit remembers whether you wrapped. You may not erase it and you may not discard it — but **you do not need to remember, because you can work it out again from the answer**: since $x < N$, $f = 1 \iff R \geq X$, so $[R < X] = \neg f$.

You cannot remember whether you borrowed during a subtraction — but you can look at your answer, deduce that you must have, and then honestly forget. Every block in this tower ends by recomputing its own flag from its own output; that discipline is what keeps the construction garbage-free.

Schoolbook multiplication is repeated shifted addition, and every shifted multiple is a *classically known constant*:

$$a x \bmod N = \left(\sum_{i=0}^{n-1} x_i (2^i a \bmod N) \right) \bmod N \implies \text{CMULT}(a) : |c\rangle |x\rangle |b\rangle \mapsto |c\rangle |x\rangle |b + c \cdot ax\rangle$$

The bit x_i decides whether its multiple joins the sum, so it becomes a control: the data has turned into conditioning.

IN-PLACE CONTROLLED MULTIPLIER (BEAUREGARD)

Let $\bar{a} = a^{-1} \bmod N$ (classical, extended Euclid). Then

$$c \cdot U_a = \text{CMULT}(\bar{a})^\dagger \cdot c\text{-SWAP} \cdot \text{CMULT}(a)$$

implements $|c\rangle |x\rangle |0\rangle \mapsto |c\rangle |a^c x \bmod N\rangle |0\rangle$.

Proof. $c = 1$:

$|x\rangle |0\rangle \rightarrow |x\rangle |ax\rangle \rightarrow |ax\rangle |x\rangle \rightarrow |ax\rangle |x - \bar{a}(ax)\rangle = |ax\rangle |0\rangle$, since

$\bar{a}a \equiv 1$. $c = 0$: every factor is the identity. **Both branches leave the scratch in $|0\rangle$, unentangled.** $\square$

```
def c_mult_acc(qc, c, A, x, y, sign, flag, N):
    for i in range(len(x)):
        c_add_mod(qc, [c, x[i]], (A << i) % N,
                  y, sign, flag, N)
```

```
def c_ua(qc, c, A, x, y, sign, flag, N):
    c_mult_acc(qc, c, A % N, x, y, sign, flag, N)
    for i in range(len(x)):
        qc.cswap(c, x[i], y[i])
    B = pow(A, -1, N) # classical inverse
    c_mult_acc(qc, c, (-B) % N, x, y, sign, flag, N)
```

Note what is *not* here: no circuit is reversed. The adjoint of “add $\bar{a}x$ ” is “add $-\bar{a}x$ ” — the dagger becomes a minus sign on a classical number.

```
def order_circuit(A, N, t=None):
    n = math.ceil(math.log2(N))
    t = t or 2 * n
    ctr, tgt = QuantumRegister(t), QuantumRegister(n)
    anc, sf = QuantumRegister(n), QuantumRegister(2)
    out = ClassicalRegister(t, "out")
    qc = QuantumCircuit(ctr, tgt, anc, sf, out)
    qc.h(ctr) # all exponents at once
    qc.x(tgt[0]) # work register := |1>
    for i in range(t): # the ladder: t rungs
        c_uq(qc, ctr[i], pow(A, 2**i, N),
            tgt, anc, sf[0], sf[1], N)
    qc.append(qft(t).inverse(), ctr)
    qc.measure(ctr, out)
    return qc
```

$4n+2$ qubits. At $N = 15$: $8 + 4 + 4 + 2 = 18$. Measured: peaks at exactly $\{0, 64, 128, 192\}$, all 4096 shots. $N = 21$: 22 qubits, depth $\approx 23,000$, $r = 6$.

The ladder is not one circuit

Every rung uses a *different* constant, so the rungs are t structurally different circuits that happen to sit side by side. **The compiled circuit is specific to a** , and a bad base means recompiling: $O(n^3)$ classical bit operations — **50 ms at 2048 bits, against days of quantum runtime**. Expected number of distinct circuits: about two.

The tax

At $n = 2048$:

$\approx 2.6 \times 10^9$	vs	$\approx 10^5$
Toffoli, modular exponentiation		gates, approximate QFT

The Fourier layer — which carries all the conceptual content — is 0.003% of the circuit.

Everything that determines whether RSA falls is the cost of multiplying two 2048-bit numbers, reversibly, in superposition.

THE CONTROL LEMMA

Let $V = G_Q \cdots G_1$ and let V_S replace G_j by $c\text{-}G_j$ for $j \in S$. Since $V_S = |0\rangle\langle 0| \otimes R_S + |1\rangle\langle 1| \otimes V$ with $R_S = \prod_{j \notin S} G_j$,

$$V_S = c\text{-}V \text{ on } W \iff R_S = \mathbb{I} \text{ on } W.$$

A gate needs a control only if leaving it uncontrolled would fail to cancel out.

Two patterns exhaust the uses. **Conjugation:** if $V = ABA^\dagger$, control B alone — a conjugating layer never needs the control. **Compensating residue:** in the seven-block adder only blocks (1), (5), (7) *need* the outer control; with the control off, (2)(3)(4)(6) conspire to do nothing. (An equivalent placement controls (1) and upgrades (6)'s CNOT to a Toffoli — **both are correct**; only the residue bookkeeping changes. The listing on the Level-2 slide uses the second.)

And the cheapest trick of all: a Toffoli that is later uncomputed — a *temporary logical AND* — costs $4T$ for the compute/uncompute *pair* instead of $7T$ each, because the uncomputation is an X -basis measurement plus a classically conditioned CZ, which is Clifford. By the no-garbage rule nearly every Toffoli in the tower is of exactly this kind: **this is not a corner case, it is the common case.**

Measured, $N = 15$, {CNOT, U }

	every gate ctrl'd	as designed
modular adder	2,634 CNOT	206
full $c\text{-}U_a$	20,892 CNOT	1,748

An order of magnitude — $12\times$ — from one lemma.

Windowing (Gidney)

Group w control qubits into one **address** and look the product up in a table of 2^w precomputed constants. A lookup by *unary iteration* costs $2(L-1)$ Toffoli — an order of magnitude below an addition.

Measured T -count, plain vs. temporary-AND lookup:
 $L=4$: $42 \rightarrow 12$; $L=32$: $434 \rightarrow 124$ — **exactly $3.50\times$ throughout.**

One counting qubit: $4n+2 \rightarrow 2n+3$

The counting register is the largest in the circuit and it is *barely used*: each qubit is Hadamarded, controls one rung, and is measured. And in the inverse QFT **every controlled rotation is controlled by a qubit about to be measured anyway. So measure it first and make the control classical** (Griffiths–Niu; Mosca–Ekert; Parker–Plenio; Beauregard).

```
for i in range(t):
    qc.h(ctr[0])
    c_ua(qc, ctr[0], pow(A, 2**(t-1-i), N), ...)
    for j in range(i):          # semiclassical QFT
        with qc.if_test((out[j], 1)):
            qc.p(-math.pi / 2**(i-j), ctr[0])
    qc.h(ctr[0]); qc.measure(ctr[0], out[i])
    with qc.if_test((out[i], 1)):
        qc.x(ctr[0])           # reset for next rung
```

$N = 15$: 11 qubits vs 18. The quantum inverse QFT has disappeared entirely, traded for mid-circuit measurement and feed-forward.

Honest verdict (EUROCRYPT 2026): “Shor likely remains the best candidate for the first quantum factorization of large integers” — Regev’s space cost is the obstacle.

Fewer gates. *Coset representation* (Zalka; Gidney–Ekerå): store $|k\rangle = 2^{-c/2} \sum_j |jN + k\rangle$ so a *plain, non-modular* addition already lands in the right class. **Level 2 collapses into Level 1**, and the sign and flag qubits go with it.

Fewer qubits. Häner–Roetteler–Svore: $2n+2$, purely Toffoli-based. Chevalignard–Fouque–Schrottenloher (CRYPTO 2025): $n/2 + o(n)$ via approximate residue arithmetic — 1,730 logical qubits for RSA-2048.

Fewer runs. Ekerå–Håstad: the exponent is $2n$ bits in Shor and **slightly more than $n/2$** in theirs — $\approx 4\times$ less modular exponentiation.

A different algorithm. Regev (2023): $\tilde{O}(n^{3/2})$ gates, $\sqrt{n} + 4$ runs. Ragavan–Vaikuntanathan cut its space to $O(n \log n)$ using *Fibonacci numbers in the exponent*.

The twist: elliptic curves

Factoring

Group $(\mathbb{Z}/N)^\times$. The oracle **multiplies by a classical constant** — and multiplication by a constant is cheap: shifted additions.

$$\lambda = \frac{y_1 - y_2}{x_1 - x_2}, \quad x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_2 - x_3) - y_2$$

Three subtractions, one squaring, one multiplication — **and one division**.

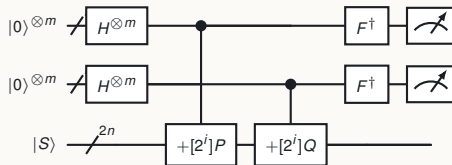
The tax

Group $E(\mathbb{F}_p)$. The oracle **adds a classical point** — and the chord-and-tangent law needs a **division**, i.e. a modular inverse *of a quantum value*.

Moral

That single fact is the whole story of this part: **an inversion costs an order of magnitude more than the multiplication it replaces**, and every optimisation in the elliptic-curve literature is an attempt to make it cheaper, do it less often, or avoid it altogether.

Everything *below* the inversion is the factoring tower with the modulus renamed. Everything *above* it — Hadamards, controlled rungs, inverse QFT — is unchanged.



$$\frac{1}{q} \sum_{u,v} |u\rangle |v\rangle |S\rangle \mapsto \frac{1}{q} \sum_{u,v} |u\rangle |v\rangle |S + [u]P + [v]Q\rangle = \frac{1}{q} \sum_{u,v} |u\rangle |v\rangle |S + \underbrace{[u + kv]}_w P\rangle$$

the last equality is the **only** place $Q_{\text{pub}} = [k]P$ is used.

Group by w : the third register depends on (u, v) *only* through $w = u + kv \bmod r$, so the state is a sum of **orthogonal** branches. It is never measured and never needs to be — *it is enough to analyse one branch*.

$2m \approx 2n$ rungs, against $2n$ for factoring: **at the level of counting, nothing has changed. The cost is entirely inside the box marked $+P$.**

Why the accumulator starts at S , not at $\mathcal{O}$

Affine formulas cannot represent the point at infinity at all: there is no (x, y) for it, and λ is undefined the moment the accumulator equals $\pm$ the addend. Starting at a random constant S keeps the accumulator on an ordinary point throughout, and **shifts the computed coset without touching the interference** — S appears only inside the third register's label.

Fix a branch w . Its (u, v) set is a coset of the **period lattice**

$$L = \{(u, v) \in \mathbb{Z}^2 : u + kv \equiv 0 \pmod{r}\} = \langle (r, 0), (-k, 1) \rangle.$$

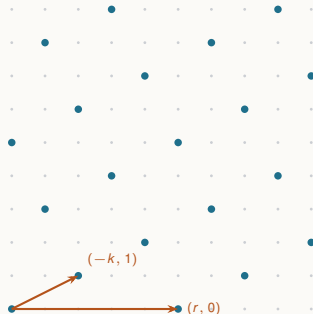
The two generators say the two things you can do without changing the answer: *take r steps in u , or one step in v and k steps back in u* . $\det L = r$, so the plane splits into exactly r cosets — one per w , as it must.

The discrete logarithm is the slope of the second generator.

Finding L is finding k ; that is the whole problem, restated.

A Fourier transform does not see L ; it sees the dual,
 $L^* = \langle \frac{1}{r}(1, k), (0, 1) \rangle$. Writing a point as $(a_1/r, a_2/r)$ and clearing denominators, membership says exactly

$$a_2 \equiv k a_1 \pmod{r} \implies k = a_2 a_1^{-1} \pmod{r}.$$



$r = 5, k = 3$: the period lattice

Moral

The congruence is not a *consequence* of the dual lattice — it *is* the dual lattice, written in coordinates. And note r is *not secret*: it is the curve's published group order. So the classical post-processing is a **division**, not a search — no continued fractions, no lcm, no repetition lottery.

Take $r \mid q$ for a moment, $q = rM$. For each v the admissible u are $u_0(v) + tr$, so

$$A(j_1, j_2) \propto \sum_v \omega^{vj_2} \omega^{u_0(v)j_1} \underbrace{\sum_{t=0}^{M-1} \omega^{trj_1}}_{(i)}.$$

Sum (i) is geometric: **M when $M \mid j_1$, exactly zero otherwise.** For the survivors $j_1 = a_1 M$, the remaining sum over v is q iff $j_2 \equiv ka_1 M$. Divide through by M : $a_2 \equiv ka_1$, **every surviving outcome has $|A|^2 = 1/r$** , and nothing else has any amplitude at all.

The clean case never actually happens

For ECDLP r is *prime* and q is a power of two, so $r \mid q$ only when $r = 2$. With $r \nmid q$, sum (i) becomes a **Dirichlet kernel** and amplitude leaks either side of $a_1 q/r$ — which is why $a_i = \text{round}(j_i r/q)$ has a rounding in it at all.

$y^2 = x^3 + 5x + 4$ over $\mathbb{F}_7$, $G = (0, 5)$ of order 5,
 $Q = [2]G$, $m = 3$

(j_1, j_2)	mass	(a_1, a_2)	$a_2 \equiv 2a_1?$	
(0, 0)	20%	(0, 0)	— ($a_1 = 0$)	
(6, 5)	10%	(4, 3)	✓	Reproduced here
(2, 3)	10%	(1, 2)	✓	
(3, 6)	10%	(2, 4)	✓	
(5, 2)	10%	(3, 1)	✓	

by direct simulation of the amplitudes.

Success rate. One run works iff a_1 is invertible: for prime r , that is $(r - 1)/r$ — and it does *not* improve with more qubits, because $a_1 = 0$ is a fixed $1/r$ of the mass. With the Dirichlet blur, Ekerå's analysis gives 60% at $m = \lceil \log_2 r \rceil$ rising to 88% with three extra qubits per register.

L	object	made of
5	the two ladders	$2m$ runs
4	$c-(+P_i)$	4 sub, 1 add, 2 div , 2 mul, 1 sqr
3	$x^{-1} \bmod p$	$2n$ Kaliski rounds
3'	$x \cdot y \bmod p$	n doublings + n additions
2	$\pm \bmod p$	plain add/sub + one reduction
1	ADD	CDKM chain / AND chain
0	AND	$4T$ compute , $0T$ uncompute

Level 3 is new and it is the expensive one.

A quantum circuit cannot branch

The binary extended Euclidean algorithm branches four ways per step. **Every branch must run on every input, on a superposition.** Kaliski's reformulation makes all four cases the same straight line — run $2n$ times, unconditionally:

1. if v odd and (u even or $u > v$): swap; **remember it**
2. if u, v both odd: $v \leftarrow u$; $s \leftarrow r$
3. $v \leftarrow v/2$; $r \leftarrow r \times 2$
4. if it swapped: swap back

Three decision bits per round **must survive or the round is not reversible** — $6n$ qubits of per-round record, **the purest which-path record in the circuit**, cleared only by running the whole inversion backwards.

The eleven-step point addition: how λ gets erased

$$x \leftarrow x_2 \rightarrow y \leftarrow qy_2 \rightarrow \lambda \oplus = qy/x \rightarrow y \oplus = x\lambda (=0) \rightarrow x \leftarrow 3x_2 \rightarrow x \leftarrow \lambda^2 \rightarrow y \leftarrow x\lambda \rightarrow \lambda \oplus = qy/x (=0) \rightarrow \dots$$

Step 8 is the same circuit as step 3, and it clears λ : by then $x = x_2 - x_3$ and $y = \lambda(x_2 - x_3)$, so y/x is again λ . **The whole eleven-step arrangement exists to make that identity true** — without it λ would be n qubits of garbage per addition.

$y^2 = x^3 + 5x + 4 \bmod 7, \text{ ord}(G) = 5$

ECDLP circuit	qubits	gates	depth
real arithmetic	69	105,339	70,991
permutation oracle	12	603	282
+ semiclassical QFT	7	619	287

**69 qubits and 105,339 gates
to compute a discrete logarithm
in a group of order five.**

Two *structural* answers to the inversion — the Euclidean “dialog” (record the decision bits, replay them; 54% off) and Jacobian projective coordinates (inversion-free; 45% off, at $3n$ qubits of irreducible garbage per addition) — **replace each other rather than compose.**
Every published construction picks a side.

The arithmetic is enormous, and the counting-level tricks are the only reason cryptographic sizes are discussed at all. At $n = 256$:

windowing	514 point additions $\rightarrow$ 34 ($15\times$), by looking P_i up in a table of 2^{16} pre-computed points
zig-zag	projective garbage 24,576 $\rightarrow$ 6,144 qubits, by arranging additions in runs and walking back
semiclassical	both counting registers collapse to one recycled qubit

None of these touches an arithmetic circuit, which is why they compose cleanly with everything below — and why they dominate. An optimisation that halves a point addition is worth half as much as one that runs fifteen times fewer of them.

What it costs, honestly

year	work	physical qubits	runtime	assumptions
2019	Gidney–Ekerå	20 million	8 hours	surface code, $p = 10^{-3}$, $1\ \mu\text{s}$ cycle, planar
2021	Gouzien–Sangouard	13,436	177 days	+ multimode quantum memory
2025	Gidney	< 1 million	< 1 week	+ approximate residues, yoked codes, cultivation
2026	Webster et al. (Iceberg)	$< 100,000$	~ 1 month	qLDPC — <i>not planar connectivity</i>
2026	Bluvstein et al.	$\sim 11,000\text{--}100,000$	days–months	<i>neutral atoms, 1 ms cycle</i>

The logical layer, which moves more slowly

Gidney–Ekerå’s abstract circuit: $3n + 0.002\ n \lg n$ logical qubits and $0.3\ n^3 + 0.0005\ n^3 \lg n$ Toffolis. At $n = 2048$: $\approx 6,190$ **logical qubits**, $\approx 2.6 \times 10^9$ **Toffolis**.
Gidney 2025: 6.5×10^9 Toffolis, $\approx 1,400$ peak logical qubits, 9.2 expected shots.

Read the assumptions, not the headline

20M/8h $\rightarrow$ $< 1\text{M}/< 1$ week is a **space-for-time trade**, not a pure win: the qubit count fell $\sim 20\times$ and the wall-clock rose $\sim 20\times$, so **the spacetime volume is essentially unchanged**.

The 2026 numbers buy their qubit reduction with **non-planar connectivity** or a **clock 1000 times slower**. The comparisons are not like-for-like. Say so out loud.

At equivalent classical security (≈ 128 -bit)

	logical qubits	Toffolis
RSA-3072 (HRS)	6,146	1.86×10^{13}
ECDLP 256-bit (RNSL 2017)	2,330	1.26×10^{11}
ratio	$2.6\times$ fewer	148 $\times$ fewer

The 256-bit ECDLP trajectory

	qubits	Toffolis
Roetteler et al. 2017	2,330	1.26×10^{11}
Haner et al. 2020	2,124	—
Litinski 2023	$\sim 3,000$	$(44 + 65/k) \times 10^6$
Babbush et al. 2026	$\leq 1,200$	$\leq 9 \times 10^7$
Schrottenloher 2026	1,192	$\approx 7 \times 10^7$

The tax

Elliptic curves give equivalent classical security at far smaller parameters (256-bit ECC $\approx$ 3072-bit RSA). **That compactness — the reason ECC is everywhere in TLS, SSH and code signing — becomes a liability quantumly**, because quantum cost scales in the *bit-length of the group*.

Any migration plan that prioritises RSA-2048 over P-256 has the order backwards.

Moral

Three orders of magnitude in Toffoli count in nine years, from algorithmic work alone — with no new hardware at all. That is the real content of the “optimisations” story, and the reason resource estimates should be quoted with their provenance and re-checked before they go on a slide.

The compiled-oracle caveat

Nearly every “ $N = 15$ factored on hardware” demonstration uses a **compiled oracle**: knowing r in advance, one writes the permutation $y \mapsto a^{2^i} y$ as a handful of gates — and notes that for $i \geq 2$ it is the *identity*, so no gate is needed at all.

A legitimate simulation aid and an illegitimate factoring algorithm: the circuit is constructed from the answer.

Smolin, Smith & Vargo, *Nature* 2013: “**all composite numbers admit simplification of the algorithm to a circuit equivalent to flipping coins...** Valid implementations **should not make use of the answer sought.**” They “factored” a 20,000-digit number with a handful of qubits to make the point.

Isaac Chuang, on his own 2001 experiment: “*if you simplify it sufficiently, well, then you can run it, but it’s not terribly meaningful.*”

record	value
largest <i>attempted</i> on hardware	$N \leq 35$
best-controlled (ion trap, 2016)	15
classical <i>simulation</i> of Shor	549,755,813,701
any quantum factoring (annealing)	8,219,999
RSA-2048	617 digits

Simulation beats the best *quantum-hardware* factorisation — the annealing record — by 16 bits. The annealing record recasts factoring as a QUBO minimisation: **no exponential speedup** — measured scaling is “absolutely and asymptotically better than random guessing but still exponential”.

The tax
Treat press claims arithmetically. “RSA-2048 factored by D-Wave” (2025) restricts itself, in its own abstract, to *a class of special integers*. The 372-qubit Schnorr/QAOA claim was refuted: it factors up to 70 bits and fails at 80.

Mosca's inequality

If X (secrecy lifetime) + Y (migration time) >
 Z (time to a CRQC),

you are already late.

Deadlines, as published. NIST IR 8547: RSA-2048 and P-256 *deprecated by 2030, disallowed by 2035*. UK NCSC: discovery by 2028, high-priority migration by 2031, everything by 2035. NSA CNSA 2.0: software signing exclusively PQC by 2030, browsers and cloud by 2033.

Standards. FIPS 203/204/205 final Aug 2024 (ML-KEM, ML-DSA, SLH-DSA); HQC selected Mar 2025 as a code-based backup; FN-DSA still in development.

The tax

The asymmetry nobody plans for. Measured across 32,011 domains in June 2026:

key exchange, hybrid ML-KEM+X25519	49.3%
post-quantum certificates	0%
banking, government, defence	0%

Cloudflare reports > 50% of human web traffic post-quantum-encrypted, up from ~3% in early 2024.

Key exchange is halfway migrated. Authentication is at zero. Harvest-now-decrypt-later is being defended against; long-lived signing keys and certificate chains are not.

And PQC protects *future* confidentiality, never past: already-recorded traffic stays exposed whatever you deploy tomorrow. Expert survey (GRI 2025, 26 respondents): probability of a cryptographically relevant machine within 10 years, 28–49%; within 20 years, 92% of respondents say $\geq 50\%$.

$$N = pq \bullet n = \lceil \log_2 N \rceil \bullet r = \text{ord}_N(a) \bullet t = 2n \bullet U_a |y\rangle = |ay \bmod N\rangle \bullet |u_S\rangle = \frac{1}{\sqrt{r}} \sum_k e^{-2\pi i s k / r} |a^k\rangle \bullet F |j\rangle = \frac{1}{\sqrt{Q}} \sum_k e^{2\pi i j k / Q} |k\rangle$$

Moral

[Abelian HSP: solved] One-dimensional irreps $\Rightarrow$ scalar-valued Fourier transform $\Rightarrow$ efficient circuit $\Rightarrow$ uniform annihilator sampling.

Order finding; discrete log over $\mathbb{F}_p^\times$ and over elliptic curves; Simon's problem. This is the *exact* cryptanalytic reach of Shor: RSA, finite-field Diffie–Hellman, DSA, ECDH and ECDSA fall together — one theorem applied five times.

The tax

Nonabelian HSP: not solved. With irreps of dimension > 1 the transform is matrix-valued; measurement returns a representation label *plus* matrix indices; reconstructing H appears to need entangled measurements across many coset states.

Graph isomorphism sits at the symmetric group. **Lattice problems reduce to the *dihedral* group**, where the best known algorithm is subexponential, not polynomial — **open since the late 1990s.**

The structural answer to “why are the replacements believed safe?”

The hardness of lattice problems **does not present as an abelian hidden subgroup**, and the one representation-theoretic route known runs into the dihedral case. **Note precisely what this does and does not claim: the belief rests on the *absence of an algorithm*, not on a proof.**

And Grover is not Shor. Symmetric primitives lose at most half their security, so AES-256 and SHA-384 survive. Saying that out loud makes the asymmetric urgency *more* credible, not less.

$$N = pq \bullet n = \lceil \log_2 N \rceil \bullet r = \text{ord}_N(a) \bullet t = 2n \bullet U_a |y\rangle = |ay \bmod N\rangle \bullet |u_S\rangle = \frac{1}{\sqrt{r}} \sum_k e^{-2\pi i k s / r} |a^k\rangle \bullet F |j\rangle = \frac{1}{\sqrt{Q}} \sum_k e^{2\pi i j k / Q} |k\rangle$$

1 The answer is spectral, and the input is free.

No residue mod N knows r locally, but the operator $y \mapsto ay$ knows it globally: its spectrum is the regular r -gon $\{e^{2\pi i s/r}\}$. And $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$ costs one X gate — **you never have to prepare an eigenvector you could not write down.**

2 Branches interfere only if they agree on every register.

One dirty scratch bit multiplies every cross term by $\langle g_{x'} | g_x \rangle = 0$ and turns the answer into a flat histogram — **a perfectly unitary circuit that computes nothing.** Half of every arithmetic circuit exists to undo the other half. **That is not waste; it is the price of the interference.**

3 At $n = 2048$ the Fourier layer is 0.003% of the circuit.

Everything that determines whether RSA falls is the cost of multiplying two 2048-bit numbers, reversibly, in superposition. **And by the same accounting, P-256 is the cheaper target, not the safer one.**

Nothing in either circuit ever computes the secret.

The circuit changes basis until the secret is the label of a measurement outcome.

Algorithms.

- Is Regev's $\tilde{O}(n^{3/2})$ circuit ever the right choice in practice, or does its space cost keep Shor in front?
- The dihedral HSP — subexponential since 2003. Anything better would change the PQC picture.
- How far can qubit counts fall? $n/2 + o(n)$ for factoring; $\approx 1,200$ logical for P-256, down from 2,330 in 2017. Where is the floor?

Engineering, and honesty about it.

- Every 2026 headline buys qubits with connectivity or clock speed. **What is the right single figure of merit?** (Spacetime volume is the current answer.)
- The gap between $N \leq 35$ on hardware and 617 digits is not closed by any curve anyone has drawn.
- **Nobody has run an *uncompiled* Shor circuit at a size where it means anything.** That, not the qubit count, is the demonstration to watch for.

Moral

The feature that makes Shor work and the feature that makes it hard to build are **the same one**. Stray coupling to an environment, $\sum_x \psi_x |x\rangle |E_0\rangle \rightarrow \sum_x \psi_x |x\rangle |E_x\rangle$, is the master formula with the environment as the register you failed to uncompute — and complete decoherence flattens the output to exactly 2^{-t} .

Decoherence is not noise added on top of a good answer; it is the continuous erasure of the only feature the algorithm produces. Fault tolerance is the price of using a phase as a data register.

Thank you

**tensor products → controlled operations → eigenvectors and kickback
→ superpositions of eigenvectors → entanglement → which-path records
→ interference and uncomputation → phase estimation → Shor**

Backup

for the questions you can already hear coming

How this deck is timed

Three paths through the same material

path	frames	length
core (clocked)	35	52 min
+ optional ($\asymp$)	49	≈ 72 min
lean (see below)	30	≈ 45 min

Wall-clock times in the frame titles track the **core path**: skip every $\asymp$ frame and the clock stays honest.

Two ending points. *A* — “ECC is not the safer half” at 47:00, then straight to “Three things”. *B* — the full landing at 50:00. Choose live.

The persistent strip along the bottom is deliberate: it is the slide-talk substitute for a side board that stays up all hour. Point at it two or three times.

The three droppable modules, in the order to drop them

1. **Arithmetic depth** — Draper’s adder, the damage law, and (if desperate) Levels 3–4. Costs you the “0.003%” punchline only if you also drop Level 5; don’t.
2. **Angles** — “One theorem, four appearances”, the character/HSP slide, “Operators are basis-free”. The mathematicians will miss these; the physicists will not.
3. **Elliptic-curve interior** — the amplitude derivation and the Kaliski/point-addition slide. Keep the two-lattice slide and the “69 qubits for a group of order five” number: they carry the act.

“Isn’t this just the uncertainty principle?”

Yes — twice. And neither time is the one people expect. Nothing here involves position and momentum, and nothing involves a measurement disturbing anything: the circuit is *unitary* from end to end, and nothing is measured until the last instant.

1. The Dirichlet peak is a bandwidth limit

You are estimating a frequency from a window of 2^t samples, and **a finite window cannot resolve a frequency exactly** — the theorem that says a click has no pitch and a pure tone has no instant.

So the peak has width ~ 1 bin, and **“each extra counting qubit buys one bit”** is the aperture–resolution trade. The $4/\pi^2$ is what a *rectangular* window costs you in leakage — and, as in signal processing, better window functions do help.

2. The garbage law *is* complementarity

$\langle g_{x'} | g_x \rangle$ multiplying every cross term is **which-path information**. For two branches the fringe visibility V and the distinguishability D obey

$$V^2 + D^2 \leq 1, \quad V = | \langle g_{x'} | g_x \rangle |$$

(Greenberger–Yasin; Englert). **Uncomputation is precisely the operation that drives $D \rightarrow 0$ and restores $V \rightarrow 1$:** a genuine uncertainty relation, and a trade between two incompatible facts about one state.

Moral

Both links run through the same place — **Fourier duality: sharp in one basis means spread in the other**. That is a theorem about functions, not a fact about hardware, which is why signal processing had it before quantum mechanics did. **“Uncertainty” here is a constraint on what a state can be, not a limit on how carefully you are allowed to look.**

Phase estimation is a spectrometer

Define the spectral measure of the work state, $\mu_\phi = \sum_k |c_k|^2 \delta_{\theta_k}$ on $[0, 1)$.

SPECTRAL-MEASURE THEOREM

For any work state $|\phi\rangle$, $\Pr[y] = \int_0^1 |A_y(\theta)|^2 d\mu_\phi(\theta) = \sum_k |c_k|^2 |A_y(\theta_k)|^2$.

Feed it a state, it reports which eigenfrequencies of U that state contains, with weights $|c_k|^2$ and resolution $1/Q$. **The three cases are three shapes of μ_ϕ , and nothing else distinguishes them.**

	1. eigenvector	2. superposition	3. generic (Haar-random in $\mathbb{C}^D$)
spectral measure	one line δ_θ	a few atoms	$\approx$ uniform on D atoms
overlap z_Δ	$e^{2\pi i \Delta \theta}$, $ z = 1$	$\sum_k c_k ^2 e^{2\pi i \Delta \theta_k}$	decays, $ z_\Delta \sim D^{-1/2}$
the state	product, rank 1	entangled, rank = $\#\{c_k \neq 0\}$	entangled, rank $\approx D$
$\Pr[y]$	$ A_y(\theta) ^2$	$\sum_k c_k ^2 A_y(\theta_k) ^2$	near-uniform once $D \gtrsim Q$
Bloch length $ z $	1	< 1	$\rightarrow 0$
outcome	θ to t bits	one random θ_k , w.p. $ c_k ^2$	no usable information
entanglement needed?	no — bilinearity alone	yes — orthogonality kills the cross terms	yes, and it is the obstruction

Moral

Entanglement rises $0 \rightarrow$ moderate $\rightarrow$ maximal while **usefulness falls** high $\rightarrow$ high $\rightarrow$ zero. **Entanglement is not the resource.** What matters is the *structure* of the correlation. Shor lives in Case 2 by necessity, and the entire design effort is spent keeping it there.

The geometry: entanglement as a shortened arrow

Run one controlled- U on $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |\psi\rangle$ and trace out the work register:

$$\rho_c = \frac{1}{2} \begin{pmatrix} 1 & \bar{\chi} \\ \chi & 1 \end{pmatrix}, \quad \chi = \langle \psi | U | \psi \rangle = \sum_s |c_s|^2 e^{2\pi i \varphi_s}.$$

$|\chi|$ — the control's transverse Bloch length — is **the population-weighted centroid of the eigenphase dials**. One number, three meanings at once: the Ramsey fringe visibility ($p(0) = \frac{1}{2}(1 + \text{Re } \chi)$), the coherence surviving entanglement, and a convex combination of points on the unit circle. Purity = $\frac{1}{2}(1 + |\chi|^2)$.

Work = $|1\rangle$: $\chi = \frac{1}{r} \sum_s e^{2\pi i s/r} = 0$ — the centroid of the r -gon is its centre. **The control qubit alone is maximally mixed: no single qubit knows anything.**

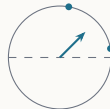
Moral

The eigenphase s/r lives **entirely in the correlations**, and the **inverse QFT — a change of basis on the control register alone — converts exactly those correlations into readable populations**. The pre-measurement state has **Schmidt rank exactly r** , so the entanglement entropy never exceeds $\log_2 r$ bits: **thin and structured, a single bond of dimension r , not an amorphous tangle**.



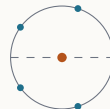
(i) eigenvector

$$|\chi| = 1$$



(ii) two modes

$$0 < |\chi| < 1$$



(iii) work = $|1\rangle$

$$\chi = 0$$

The same conclusion in Reading A: the comb

Condition instead on the *work* register. Outcome $|a^{x_0}\rangle$ leaves the counting register in

$$\frac{1}{\sqrt{K}} \sum_{j=0}^{K-1} |x_0 + jr\rangle, \quad K \approx 2^t/r$$

— a perfectly coherent **comb** of spacing r and unknown offset x_0 . (F4: conditioning destroys only which-branch information, not coherence *within* the branch.)

Moral

Under $F^\dagger$ the offset contributes only the phase $e^{-2\pi i \ell x_0 / 2^t}$ per outcome — **modulus one, gone at the squaring**. So the distribution peaks at multiples of $2^t/r$ *regardless of* x_0 : **the unknown offset is harmless**, and the output distribution is identical to Reading B's.

Which register carries “the answer” is, again, a matter of description.

With $Q = qr + b$ the fibres have *unequal* sizes differing by one ($N = 21$, $a = 11$, $Q = 1024$: 171×4 , 170×2). The truncated geometric series is why the peaks acquire width — **spectral leakage**.

The measurement that need not happen

Suppose someone measured the work register in the eigenbasis *before* the circuit ran. That measurement is in an eigenbasis of U_a , so it **commutes with every rung** — it can be slid to the end, where it manifestly cannot affect the counting statistics.

Hence it need not be performed at all, and the work register is never measured, never uncomputed, and may simply be discarded.

The two-sided rule: which records to keep

The master formula treats *both* unmeasured registers identically, which means the design task is to arrange **opposite** values for them:

$$\langle f(x') | f(x) \rangle = \delta_{c,c'} \quad \text{a record you must protect,} \quad \langle g(x') | g(x) \rangle \equiv 1 \quad \text{a record you must destroy.}$$

Clear the scratch; keep the accumulator

Clear every record that distinguishes branches you *want* to interfere. The r same-coset branches must stay indistinguishable, or the character sum never fires.

Keep the record that distinguishes branches you *must not* let interfere. The accumulator's orthogonality is what stops different cosets from blurring together — and it is why measuring it is optional.

A register may survive to the transform iff its contents are a function of the coset label alone. Apply “always uncompute” uniformly and you restore one state on every branch, making the state a product, whose inverse transform is $|0\rangle$: **you measure $(0, 0)$ with certainty.** Apply “records are harmless if unread” and you get uniform noise.

Nothing in the syntax of a reversible circuit marks which register is which. It is a property of the algorithm.

Why scratch varies *within* a coset. At $r = 7$, $k = 4$ the branches $(0, 0)$ and $(3, 1)$ lie in the same coset, yet they agree at the start and at the end **and nowhere in between**: every intermediate accumulator differs, so every slope differs. **Convergence is a property of the final value only** — which is why the garbage is enormous and every bit of it separates branches required to add coherently.

The CDKM adder, traced: $a = 5$, $b = 3$, $n = 3$

after	c_0	b_0	a_0	b_1	a_1	b_2	a_2	z
start	0	1	1	1	0	0	1	0
MAJ(c_0, b_0, a_0)	1	0	1	1	0	0	1	0
MAJ(a_0, b_1, a_1)	1	0	1	1	1	0	1	0
MAJ(a_1, b_2, a_2)	1	0	1	1	0	1	1	0
CNOT $a_2 \rightarrow z$	1	0	1	1	0	1	1	1
UMA(a_1, b_2, a_2)	1	0	1	1	1	0	1	1
UMA(a_0, b_1, a_1)	1	0	1	0	0	0	1	1
UMA(c_0, b_0, a_0)	0	0	1	0	0	0	1	1

Moral

Read the b column downwards: it starts as $011 = 3$ and ends as 000 with $z = 1$ — that is $1000_2 = 8$. The a wires end at $1, 0, 1 = 5$, exactly as they started, and c_0 is back to 0. **Nothing is left dirty.**

Why the single CNOT sits exactly there

The only wire holding a clean, unmodified carry at any moment is the topmost one, a_2 , immediately after the last MAJ. That is exactly where the CNOT is. The placement is not arbitrary: it is **the one instant at which the final carry is sitting in the clear.**

And why the wires look scrambled: the order is $c_0, b_0, a_0, b_1, a_1, \dots, z$ — alternating. **It means nothing mathematically. It is a seating plan:** interleave them and each block's three qubits are always adjacent. Re-order the wires and the circuit is identical, only uglier.

Backup code: the QFT, and the ripple-carry backend

```
def qft(n, approx=0):
    """|j> -> 2-(n/2) sum_k ei(2 pi i j k / 2n) |k>.
    approx > 0 drops rotations below pi/2approx."""
    qc = QuantumCircuit(n, name="qft")
    for j in reversed(range(n)):
        qc.h(j)
        for k in range(j):
            if approx and (j - k) > approx:
                continue
            qc.cp(np.pi / 2**(j - k), k, j)
    for i in range(n // 2):
        qc.swap(i, n - 1 - i)
    return qc.to_gate()
```

$n(n+1)/2$ gates plus $\lfloor n/2 \rfloor$ swaps. With the cutoff $d = \lceil \log_2 n \rceil + 2$ the count drops to $O(n \log n)$: measured 144 $\rightarrow$ 99 gates at $n = 16$. It is one

continue, so there is no reason not to have it.

Do not test the QFT by eye. Build its unitary and compare with the definition. Drop the swap layer, or reverse the loop order, and you get a QFT of the *bit-reversed* integer — still unitary, still looks fine, and it will make every addition below wrong.

```
def rc_add(qc, x, y, carry, ctrls=()):
    """|x>|y> -> |x>|(y + x) mod 2n."""
    n = len(y)
    _maj(qc, carry, y[0], x[0], ctrls)
    for i in range(n - 1):
        _maj(qc, x[i], y[i+1], x[i+1], ctrls)
    for i in range(n - 1, 0, -1):
        _uma(qc, x[i-1], y[i], x[i], ctrls)
    _uma(qc, carry, y[0], x[0], ctrls)
```

Control telescoping

Of MAJ's and UMA's six gates, **exactly two accept the outer control** — both the CNOT targeting *b*. With the control off, MAJ leaves two gates and UMA leaves *the same two in the opposite order*: each other's inverse. Since the UMA chain retraces the MAJ chain, **the residues cancel pairwise from the inside out and the whole adder telescopes to the identity**. A controlled n -bit addition costs $2n$ controlled gates, not $6n$.

Backup code: windowing, and the temporary AND

```
def _walk(qc, ctrl, addr, out, table, offset, anc):
    """out ^= table[offset + value(addr)], on ctrl."""
    if not addr:
        T = table[offset] if offset < len(table) else 0
        for b in range(len(out)):
            if (T >> b) & 1:
                qc.cx(ctrl, out[b])    # leaf: CNOTs
        return
    a, rest = addr[-1], addr[:-1]    # split on the MSB
    half, node = 1 << len(rest), anc[0]
    qc.ccx(ctrl, a, node)             # ctrl AND a
    _walk(qc, node, rest, out, table, offset+half, anc[1:])
    qc.cx(ctrl, node)                 # ctrl AND NOT a
    _walk(qc, node, rest, out, table, offset, anc[1:])
    qc.cx(ctrl, node)
    qc.ccx(ctrl, a, node)             # ... and clear it
```

Unary iteration: $2(L - 1)$ Toffoli for an L -entry table, and only $\log_2 L$ ancillas — **not** L .

```
def and_compute(qc, a, b, t):
    # t starts |0>; leaves t = a AND b, 4 T gates
    qc.h(t); qc.t(t)
    qc.cx(b, t); qc.tdg(t)
    qc.cx(a, t); qc.t(t)
    qc.cx(b, t); qc.tdg(t)
    qc.h(t); qc.sdg(t)

def and_uncompute(qc, a, b, t, cbit):
    # erase for free: X-basis measure + Clifford
    qc.h(t)
    qc.measure(t, cbit)
    with qc.if_test((cbit, 1)):
        qc.cz(a, b)
        qc.x(t)    # ancilla back to |0>
```

The last gate is $S^\dagger$, not S

Three plausible variants of this circuit differ only in that phase and **are all wrong** — which a truth table alone will not catch. **The check has to include the phase.**

How to know your implementation is right

A wrong Shor circuit does not return a wrong answer

It returns a **flat histogram** — and since the classical post-processing verifies its own candidates, it will happily print the correct factors from pure noise. **Testing the output is therefore worth almost nothing.** Four other things are worth a great deal.

1. **Assert the ancillas are clean** — measure the scratch alongside the answer and assert $|0 \dots 0\rangle$, *at the instruction where each is freed*, not merely at the end. That is what catches a circuit computing the right value and leaving a dirty flag behind.
2. **Assert the peaks, not the factors.** “Did `find_factor` return 3 or 5?” can succeed by chance on a uniform distribution — exactly what a garbage bug produces. Assert the *distribution*.
3. **Exact simulation at full width.** Every arithmetic circuit here is a permutation of basis states, so pushing one basis state through is exact in $O(\text{gates})$ — and scales to hundreds of qubits where a statevector dies at thirty. *Sound only for the phase-free part of the circuit.*
4. **Statevector cross-checks on small instances**, looking for **leaked amplitude** — the statement that ancillas are unentangled, which basis-state testing cannot see. A stray diagonal gate appears here as a phase, and phases are exactly what Shor depends on.

A real example of why (2) matters: a post-processing search radius set too generously once enumerated the entire group — *every* candidate “verified”, on every input, including noise. **Invisible in the pass/fail of an end-to-end test; obvious the moment the vote shares were printed next to the uniform baseline.**

Resource estimates: provenance and caveats

- **Toffoli counts are not comparable across papers.** Some count CCX, CCZ and AND gates together and *exclude* measurement-based AND[†] (no Toffoli, no T); others count every three-qubit operation performed. **The two differ by nearly $2\times$ on Gidney-style circuits** — most of the modern literature. Quoting the wrong one makes every optimisation look half as good as claimed.
- **T -counts and Toffoli counts are different columns.** A Toffoli is $7T$ naively, $4T$ as a temporary AND. Do not put Häner et al.'s T -counts and Roetteler et al.'s Toffoli counts in the same column.
- **Physical-qubit headlines encode a connectivity and a clock.** 2019: planar, $1\ \mu\text{s}$. 2026 qLDPC: block-local, $1\ \mu\text{s}$. 2026 neutral atoms: reconfigurable, **1 ms**. The qubit-count reductions are partly a change of hardware model.
- **Gidney 2025 is still a preprint** (arXiv:2505.15917); its peak logical-qubit figure is $\approx 1,400$ — check Table 5 before quoting an exact integer.
- **Roetteler et al.'s own formula** $9n + 2\lceil \log_2 n \rceil + 10$ gives 2,330 at $n = 256$; Häner et al. tabulate the same baseline as 2,338. Both are in the literature.

Moral

Resource estimates are moving targets. They are logical-qubit and Toffoli figures under specific error-correction assumptions, and they improve with every optimisation paper. **Quote them with their provenance, and re-check before putting numbers on a slide.**

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